Dynamic Partial Reconfiguration

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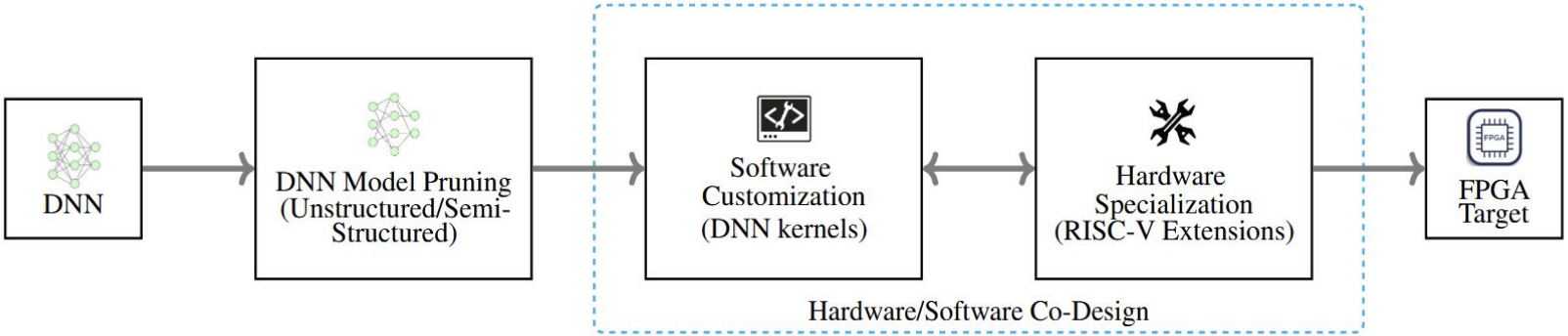
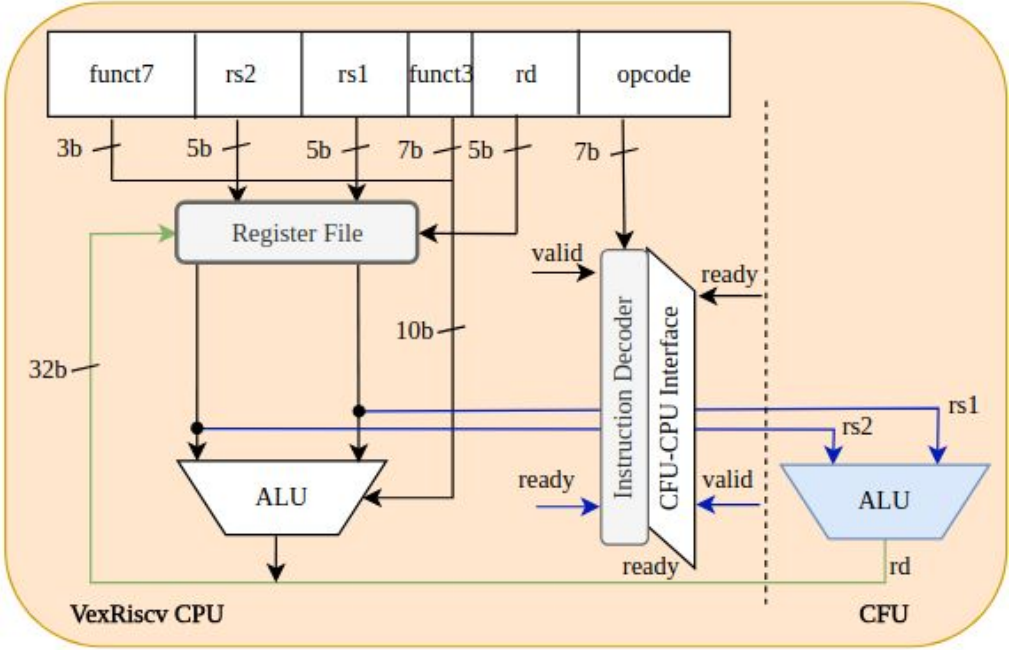
- Hardware/Software Co-Design of RISC-V Extensions for Accelerating Sparse DNNs on FPGAs

Paper's Flow

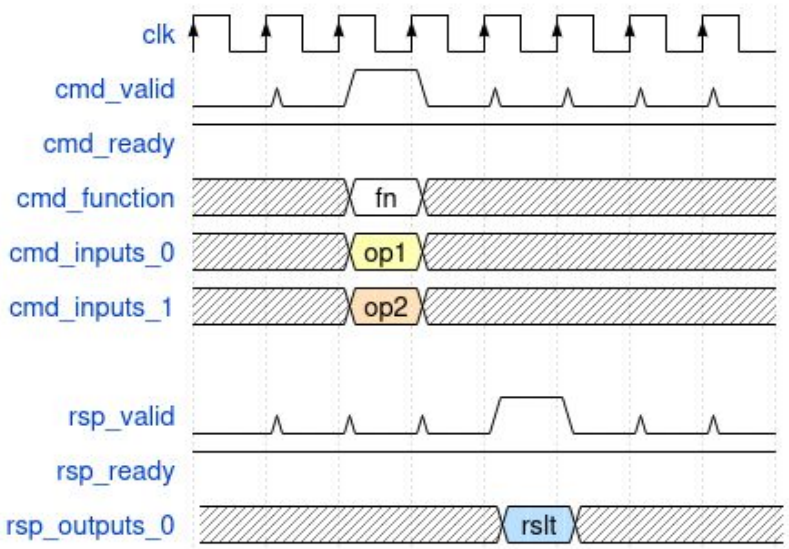

DNN
DNN Model Pruning (Unstructured/Semi-Structured)
Software Customization (DNN kernels)
Hardware Specialization (RISC-V Extensions)
FPGA Target
Hardware/Software Co-Design

CFU Architecture

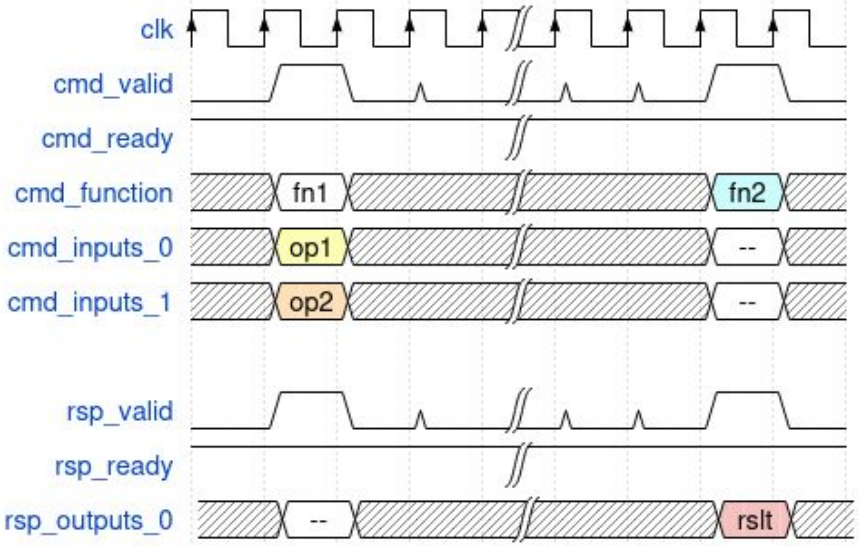


CPU-CFU Interface

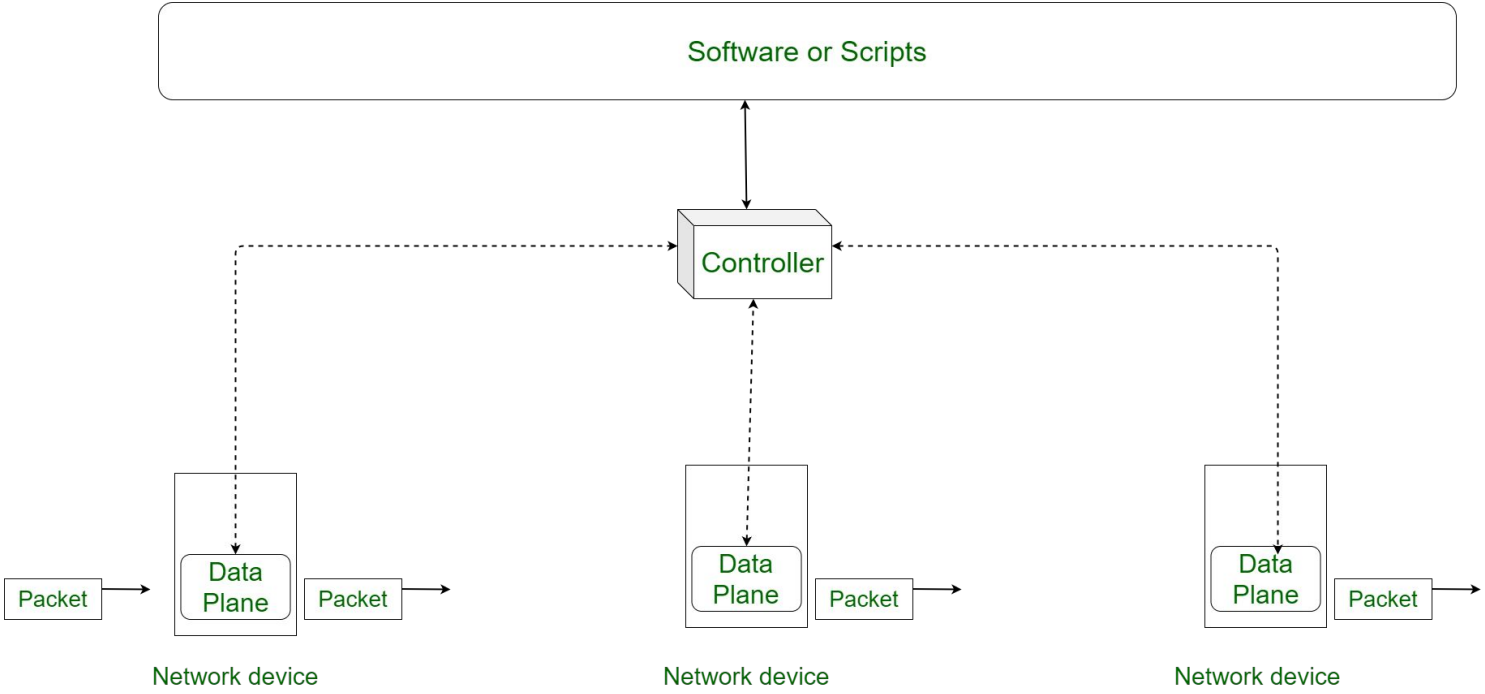
CFU with Latency



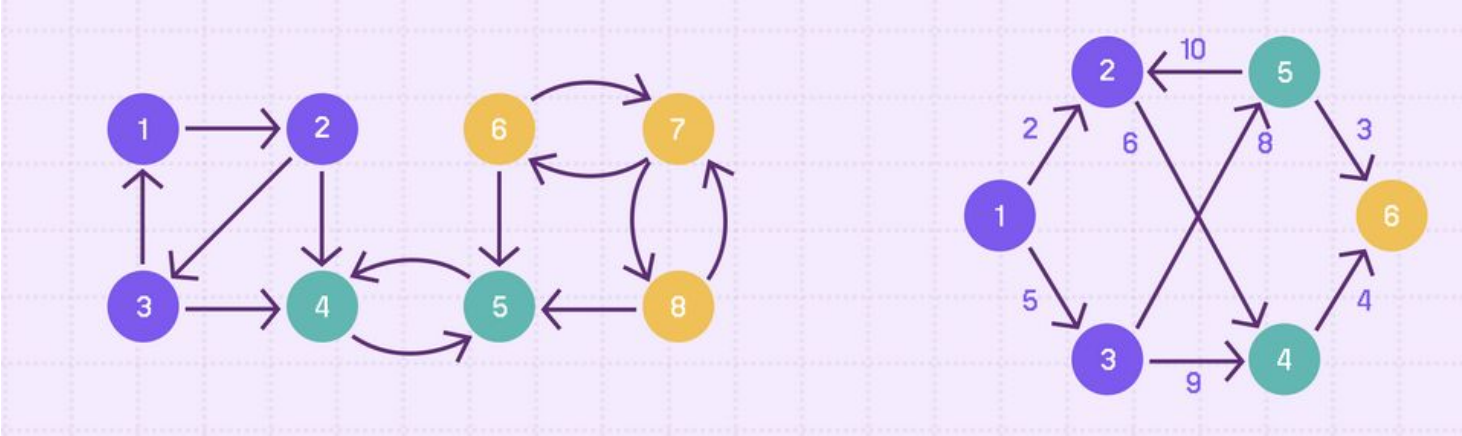
Split phase



SDNs


Software or Scripts
Controller
Packet
Data Plane
Packet
Network device
Data Plane
Packet
Network device
Data Plane
Packet
Network device

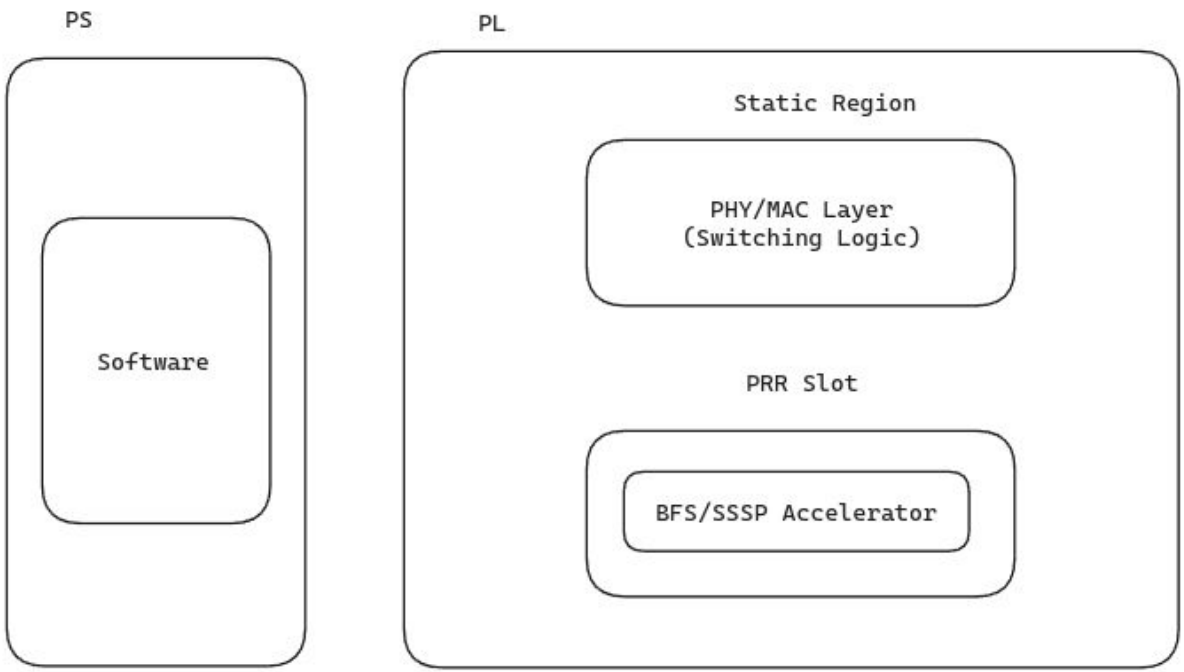
BFS vs SSSP



- Unweighted edges
- Network discovery

- Weighted edges
- Optimal path search

Architecture



Thank you